

EXPERIMENT 33

PROLOG PROGRAM TO FIND NUMBER OF VOWELS

Aim

To write a Prolog program to find the number of vowels in a given word or list of characters.

Algorithm

1. Read the input word as a list of characters.
2. Check each character one by one.
3. If the character is a, e, i, o, or u, increment the vowel count.
4. Continue until all characters are checked.
5. Display the total number of vowels.

Prolog Code

% Program to count number of vowels

vowel(a).

vowel(e).

vowel(i).

vowel(o).

vowel(u).

count_vowels([], 0).

count_vowels([H|T], Count) :-

 vowel(H),

 count_vowels(T, C),

 Count is C + 1.

count_vowels([], Count) :-

 \+ vowel(H),

 count_vowels(T, Count).

Input

count_vowels([p,r,o,g,r,a,m], Count).

Output

```
| ?- consult('C:/Users/MOSHIYA/Downloads/exp 33.pl').
compiling C:/Users/MOSHIYA/Downloads/exp 33.pl for byte code...
C:/Users/MOSHIYA/Downloads/exp 33.pl compiled, 17 lines read - 1402 bytes written, 14 ms

(15 ms) yes
| ?- count_vowels([p,r,o,g,r,a,m], Count).
Count = 2 ?
```

Result

Thus, the Prolog program to find the number of vowels was successfully implemented, and the total number of vowels in the given input was obtained.